Quadratic Residues

The concept of "square roots" in modular arithmetic.

Let's look again at quadratic equations $ax^2+bx+c=0$ and their solution for $x \in \mathbb{R}$.

$$ax^2+bx+c=0 \qquad a,b,c \in \mathbb{R},\ a \neq 0$$

$$\iff 4a^2x^2+4abx+4ac=0$$

$$\iff (2ax+b)^2-b^2+4ac=0$$

$$\iff (2ax+b)^2=b^2-4ac$$

$$\iff 2ax+b=\sqrt{b^2-4ac}$$

, provided $\sqrt{\ }$ exists.
i.e. $b^2-4ac \geq 0$

$$\iff x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}.$$

Note the two key assumptions.

- $a \neq 0$
- $b^2-4ac \geq 0$, i.e. $\sqrt{b^2-4ac}$ exists in $\mathbb{R}$.

Can we replicate this analysis for quadratic congruences modulo a prime p.

Let p be an odd prime, i.e. $p > 2$

Is there a solution $x \in \mathbb{Z}_p$

~~to $ax^2 + bx + c \equiv 0 \pmod{p}$~~

assuming $a \not\equiv 0 \pmod{p}$, i.e. $p \nmid a$.

Note this means that $4a$ is invertible modulo p, as $\gcd(4a, p) = 1$

$$ax^2 + bx + c \equiv 0 \pmod{p}.$$

$$\iff 4a^2x^2 + 4abx + 4ac \equiv 0 \quad \text{"}$$

$$\iff (2ax + b)^2 - b^2 + 4ac \equiv 0$$

$$\iff (2ax + b)^2 \equiv b^2 - 4ac \quad \text{"}$$

$$\iff 2ax + b \equiv u \quad \text{"}$$

where $u^2 \equiv b^2 - 4ac \pmod{p}$.

? provided such a $u \in \mathbb{Z}_p$ exists?

$$\iff x \equiv (-b + u)(2a)^{-1} \pmod{p}.$$

This is all valid provided

- $a \not\equiv 0 \pmod p$
- provided $b^2 - 4ac$ "has a square root" modulo p.

<u>Def 8.1</u> For an odd prime p. the element $n \in \mathbb{Z}_p$ is a quadratic residue modulo p if there are solutions to

$$x^2 \equiv n \pmod p.$$

(this at most two solutions).

If there are no solutions, n is called a quadratic non-residue modulo p.

<u>Examples</u>

$p = 7$. $\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$

but $\mathbb{Z}_7^\times = U(7) = \{1, 2, 3, 4, 5, 6\}$

residues.

$1 \equiv 1^2$, $2 \equiv 3^2$, $4 \equiv 2^2$

3, 5, 6 non-residues

$\equiv$ $\equiv$ $\equiv$

6^2 4^2 5^2

Let's switch to the computer for lots of examples and see what patterns jump out.

Some observations from the examples

- Conjecture: There are an equal number of residues and non-residues modulo p namely $\frac{p-1}{2}$
- Of course the square integers $1, 4, 9, 16, 25, \ldots$ are always residues
- Non-square integers $\neq 2$ were residues for some p, non residues for others.

A general question emerges

Given p, and $n \in \mathbb{Z}_p$.

is n a residue modulo p?

<u>Theorem 8.1</u> There are $\frac{p-1}{2}$ residues modulo p.

<u>Proof</u>. $\mathbb{Z}_p^\times = U(p) = \{1, 2, 3, \ldots, p-1\}$

Clearly the congruence classes

$$1^2, 2^2, 3^2, 4^2, \ldots, \left(\frac{p-1}{2}\right)^2 \quad (*) \pmod p$$

are quadratic residues

<u>Claim</u>: All the elements of $(*)$ are distinct mod p.

<u>Pf</u>. Suppose $n^2 \equiv m^2$ for $1 \le n, m \le \frac{p-1}{2}$

$$\iff p \mid n^2 - m^2$$

$$\iff p \mid (n-m)(n+m)$$

But $2 \le n+m \le p-1$

$$\Rightarrow p \nmid (n+m)$$

$$\iff p \mid n-m,$$ by Euclid's Lemma $p \mid ab \Rightarrow p \mid a$ or $p \mid b$

$$\iff n \equiv m \pmod p$$

$\blacksquare$

So there are at least $\frac{p-1}{2}$ residues.

Suppose j^2 is another residue for $\frac{p+1}{2} \leq j \leq p-1$

well $j \equiv -i$, for some $1 \leq i \leq \frac{p-1}{2}$

But then $j^2 \equiv (-i)^2 \equiv i^2$

So j^2 is already listed in $(*)$.

This proves there are exactly $\frac{p-1}{2}$ residues mod p, and $\frac{p-1}{2}$ non-residues.

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for square roots over $\mathbb{R}$, it's determined by number being negative or positive.

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for arithmetic mod p is there

are similar concept of positive and negative?

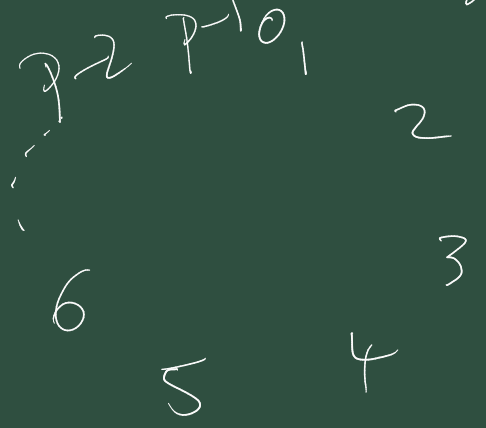
$1-3p$
$1-2p$
$1-p$

$$\mathbb{Z}_p^\times = \{1, 2, 3, \ldots, p-1\}$$

$p+1$

$2p+1$

Indeed the "model" for picturing $\mathbb{Z}_p$ is as a cycle.



There's no direct concept of pos/neg in $\mathbb{Z}_p$.

But a connection can still be made. In real arithmetic under multiplication the pos/negs behave in a certain way:

- pos × pos = pos

- neg × neg = pos
- pos × neg = neg

Is there a similar multiplicative behaviour for residues/non-residues

Examine lots of examples we can <u>conjecture</u>

1. res × res = res
2. non-res × non-res = res
3. res × non-res = non-res.

Yes these are a theorem.

<u>Proof</u>. Let p be an odd-prime.

1. Let α, β be two residues.

$$\Rightarrow \alpha \equiv a^2, \beta \equiv b^2 \pmod{p}$$

$$\Rightarrow \alpha\beta \equiv a^2 b^2$$

$$\Rightarrow \alpha\beta = (ab)^2 \pmod{p}$$

$\Rightarrow \alpha\beta$ is a residue.

(3). Let α be a residue and $\boxed{\beta \text{ be a } \underline{\text{non-residue}}}$

$\Rightarrow \alpha \equiv a^2 \pmod{p}$.

Let's assume $\alpha\beta$ is a residue

i.e. $\alpha\beta \equiv c^2$, for some $c \in \mathbb{Z}_p$.

$\Rightarrow a^2\beta \equiv c^2 \pmod{p}$.

$\Rightarrow \beta \equiv c^2(a^{-1})^2$, where a^{-1} is the inverse of a mod p.

$\equiv (ca^{-1})^2$

This contradicts the fact that β is a non-residue

Therefore, by proof-by-contradiction, $\alpha\beta$ is a non-residue.

(2) Let $\underline{\alpha, \beta \text{ both be non-residues}}$ mod p.

Let $r_1, r_2, \ldots, r_{\frac{p-1}{2}}$ be the

residues mod p. By part ③ that

αr_i, $i = 1, \dots, \frac{p-1}{2}$ are all non-residues. But these are all distinct mod p. So αr_i are <u>all</u> the non-residues mod p.

So what could $\alpha\beta$ be?

<u>If $\alpha\beta$ is a non-residue</u>

$\Longrightarrow \quad \alpha\beta = \alpha r_i$, for some i

$\Longrightarrow \quad \beta = r_i$, a residue.

But this contradicts the fact that β is a non-residue.

Therefore $\alpha\beta$ is a residue ✓

So this suggest residue/non-residue is the important concept for $\mathbb{Z}_p$.

Still faced with a fundamental question.

Is n a residue mod p?

At the moment we only have brute force calculation.

But we seek some theoretical insight.

<u>Def 8.2</u> A piece of notation.

The Legendre symbol. A kind of indicator function for residue status.

$$(n \mid p) = \begin{cases} +1 & \text{if } n \text{ is a residue mod } p \\ -1 & \text{if } n \text{ is a non-residue.} \end{cases}$$

n from $\mathbb{Z}_p^\times$.

[Some books/authors write the symbol $\left(\frac{n}{p}\right)$.

Can also extend this to define $(n \mid p) = 0$ when $p \mid n$.

Notice that our ~~multi~~ multiplicative structure res × res = res, etc. can be expressed now as saying the Legendre symbol $(m|p)$ mod p is multiplicative.

$$\boxed{\forall\, n, m \in \mathbb{Z}_p^{\times} \quad (nm|p) = (n|p)(m|p)}$$

This already will give some important insight to the general Question $(n|p) = ?$

By the F.T.A. we should think of n in terms of its prime factorization

$$n = \prod_{i=1}^{r} p_i^{a_i}$$

Assuming $n \not\equiv 0 \bmod p$ means $p \neq p_i \quad i = 1, \dots, r$

$$(n|p) = \left(\prod_{i=1}^{r} p_i^{a_i} \,\middle|\, p\right).$$

$$= \prod_{i=1}^{r} (p_i|p)^{a_i},$$ since $(n|p)$ is multiplicative.

$$= \prod_{\substack{i=1 \\ a_i \text{ odd}}}^{r} (p_i|p).$$

So in order to understand $(n|p)$ we just need to understand $(p|q)$ for primes p, q.

$$(p|q) = ? \text{ formula?}$$

Not quite a direct formula, but an intriguing insight into $(p|q)$ called <u>Quadratic</u>

reciprocity, which will give an "efficient" algorithm for calculating $(n|p)$.